Anton Vasin

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Master of Science in Software Engineering
Senior Programmer at Playrix

Technical proficiencies

Highly proficient in Modern C++.
Expert in CUDA C++, OpenGL, Apple Core Image, Java Native Interface, and WebRTC.
Experienced in Unity, C#, NodeJS, and FFMPEG.

Languages

Russian: Native
English: Advanced

Professional experience

Playrix - Remote — Sept/2023 - Present
https://playrix.com/
Senior Programmer

Technologies: C++17, Boost C++ Libraries, OpenGL ES 3.0, GLSL

Scope of my work:
- Designing the architecture of the game engine rendering system.
- Development of the image processing algorithms and shader programs.
- Designing and development of unit tests and automated tests.
- Porting the games based on the game engine to mobile (iOS, Android) and desktop (Windows, MacOS) platforms.
- Maintaining the game engine code base. Bug fixes, performance, and memory consumption optimizations.

Prequel Inc - Remote — May/2021 - Aug/2023
https://www.prequel.app/
Lead Graphics Engineer

Technologies: C++17, Boost C++ Libraries, OpenGL ES 3.0, GLSL, Metal, Apple Core Image, Directed Acyclic Graphs (DAG), Java Native Interface, Swift.

I was the lead developer of a state-of-the-art cross-platform image processing engine for photo and video editing software.

Key Achievements:
- I designed the engine architecture from scratch.
- I developed an efficient shader graph compiler.
- I implemented more than 300 image-processing filters & shaders.
- I built an efficient color management subsystem.
- I implemented Java Native Interface layers for the Android platform.
- I also implemented Swift layers for iOS & MacOS platforms.

Topcon Positioning Systems - Moscow, Russia — Jun/2019 - May/2021
https://www.topconpositioning.com/
Lead Software Engineer

Technologies: C++17, Qt, OpenGL ES 3.0, DirectX 11.

I worked on the in-house 3D engine for various mobile and desktop geodetic & survey CAD products.

Key Achievements:

- I completely overhauled the legacy part by fixing dozens of deadlocks, data races, and race-condition-induced crashes.
- I improved the concurrency model by replacing constantly conflicting mutexes with the Event Loop approach.
- I significantly optimized battery consumption on autonomous geodetic and mobile devices by significantly optimizing rendering algorithms.
- I designed and developed a new renderer based on Entity-Component-System architecture (ECS). The ECS approach improved rendering performance and decreased application response time on autonomous and mobile devices.

Competentum Group (EPAM subsidiary) - Moscow, Russia — Nov/2015 - May/2019
https://competentum.ru/
https://www.epam-group.ru/e-learning
Senior Software Developer

Technologies: C++11, C#, Unity, FFMPEG, WebRTC, Node.JS, Kurento, Docker Compose, AWS EC2, AWS S3.

I successfully finished several E-Learning web projects. The most significant was a project for Mursion, Inc. I was a lead of a team of three developers.

Key Achievements:

- I integrated the WebRTC C++ library as a peer-to-peer network system into a Unity-based game.
- I integrated the FFMPEG C library as a game video-capturing system into the game.
- I developed a game character facial animation control system based on webcam facial recognition.
- I successfully made ports of the application for Android, iOS, OSX, and Oculus Go VR Headset.

NANO Security - Bryansk, Russia — Jul/2013 - Jun/2015
https://www.nanoav.pro/
Middle Software Developer

Technologies: C++03, Boost, Microsoft COM, WinAPI, IDA Disassembler.

I started my career as a C++ developer here.

Key Achievements:

- I developed several static malware analysis modules.
- I developed various data extractors from compressed archives, file system images, installation packages, game engine bundles, etc.
- I performed reverse engineering and manual analysis of Android and Java malware.

Education

Master of Science in Software Engineering, Bryansk State Technical University